Bronchiectasis.

PMID: 7612755 | Indexed in PubMed

Bronchiectasis is a chronic suppurative respiratory disease of declining prevalence but continuing morbidity due to recurrent respiratory infections and bronchial bleeding. Literature was reviewed through a Medline search for the past 4 years. Almost all reports are retrospective case summaries. This review focuses on the clinical assessment, stressing the recognition of an impaired host and infectious contributions. High-resolution computed tomography has supplanted bronchography as a key diagnostic tool. Treatment includes prompt attention to acute bacterial infections, aerosol bronchodilators and inflammatory agents, and bronchial hygiene. The role of surgery has declined at least by the number of literature reports. Except for cures for local bronchiectasis, surgical resection is now used to palliate disease activity by removing the most affected segments and lobes.

Bronchiectasis: the 'other' obstructive lung disease.

PMID: 10418580 | Indexed in PubMed

Bronchiectasis belongs to the family of chronic obstructive lung diseases, even though it is much less common than asthma, chronic bronchitis, or emphysema. Clinical features of these entities overlap significantly. The triad of chronic cough, sputum production, and hemoptysis always should bring bronchiectasis to mind as a possible cause. Chronic airway inflammation leads to bronchial dilation and destruction, resulting in recurrent sputum overproduction and pneumonitis. Once the diagnosis is confirmed, any potential predisposing conditions should be aggressively sought. The relapsing nature of bronchiectasis can be controlled with antibiotics, chest physiotherapy, inhaled bronchodilators, proper hydration, and good nutrition. In rare circumstances, surgical resection or bilateral lung transplantation may be the only option available for improving quality of life. Prognosis is generally good but varies with the underlying syndrome.

Chronic suppurative lung disease and bronchiectasis in children and adults in Australia and New Zealand.

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Consensus recommendations for managing chronic suppurative lung disease (CSLD) and bronchiectasis, based on systematic reviews, were developed for Australian and New Zealand children and adults during a multidisciplinary workshop. The diagnosis of bronchiectasis requires a high-resolution computed tomography scan of the chest. People with symptoms of bronchiectasis, but non-diagnostic scans, have CSLD, which may progress to radiological bronchiectasis. CSLD/bronchiectasis is suspected when chronic wet cough persists beyond 8 weeks. Initial assessment requires specialist expertise. Specialist referral is also required for children who have either two or more episodes of chronic (> 4 weeks) wet cough per year that respond to antibiotics, or chest radiographic abnormalities persisting for at least 6 weeks after appropriate therapy. Intensive treatment seeks to improve symptom control, reduce frequency of acute pulmonary exacerbations, preserve lung function, and maintain a good quality of life. Antibiotic selection for acute infective episodes is based on results of lower airway culture, local antibiotic susceptibility patterns, clinical severity and patient tolerance. Patients whose condition does not respond promptly or adequately to oral antibiotics are hospitalised for more intensive treatments, including intravenous antibiotics. Ongoing treatment requires regular and coordinated primary health care and specialist review, including monitoring for complications and comorbidities. Chest physiotherapy and regular exercise should be encouraged, nutrition optimised, environmental pollutants (including tobacco smoke) avoided, and vaccines administered according to national immunisation schedules. Individualised long-term use of oral or nebulised antibiotics, corticosteroids, bronchodilators and mucoactive agents may provide a benefit, but are not recommended routinely.

Bronchiectasis in the Last Five Years: New Developments.

PMID: 27941638 | Indexed in PubMed

Bronchiectasis, a chronic lung disease characterised by cough and purulent sputum, recurrent infections, and airway damage, is associated with considerable morbidity and mortality. To date, treatment options have been limited to physiotherapy to clear sputum and antibiotics to treat acute infections. Over the last decade, there has been significant progress in understanding the epidemiology, pathophysiology, and microbiology of this disorder. Over the last five years, methods of assessing severity have been developed, the role of macrolide antibiotic therapy in reducing exacerbations cemented, and inhaled antibiotic therapies show promise in the treatment of chronic Pseudomonas aeruginosa infection. Novel therapies are currently undergoing Phase 1 and 2 trials. This review aims to address the major developments within the field of bronchiectasis over this time.

[Living with bronchiectasis].

PMID: 29143504 | Indexed in PubMed

Bronchiectasis is a chronic condition with a prevalence continuously on the rise. Bronchiectasis have a considerable impact on morbidity, healthcare utilization and quality of life. Pulmonary function tests, microbiological variables and exacerbation rate are useful in the initial and follow-up evaluation. Scores that combine those variables with chest CT findings have been established to predict hospitalizations and mortality. Assessment of health-related quality of life cannot rely on physiological variables measurement. Dedicated questionnaires are therefore needed for that purpose.

Non-cystic fibrosis bronchiectasis.

PMID: 23728208 | Indexed in PubMed

Bronchiectasis is a chronic debilitating condition with considerable phenotypic diversity. A vicious cycle of infection and inflammation exists in damaged airways with patients suffering from persistent cough, purulent sputum production, recurrent chest infections and general malaise. The associated burden of disease in terms of increased morbidity, reduced quality of life and the socioeconomic cost of long-term management is significant. Further research is essential to improve our understanding of the development and progression of this disease. This article reviews what is currently known about bronchiectasis, its pathophysiology, aetiology and management strategies.

Advances in bronchiectasis.

PMID: 31092516 | Indexed in PubMed

Bronchiectasis is a chronic inflammatory condition with a diverse aetiology including recurrent infections, genetic abnormalities, immunodeficiency and autoimmune disorders. The prevalence has increased over the past few years and this may be due to better imaging and diagnostic techniques. Management remains the emphasis for improving symptoms and reducing exacerbations. This article focuses on highlighting the latest data released since 2014 on new diagnostic techniques as well as potential future pharmacological and non-pharmacological treatment options for patients with bronchiectasis.

A review of non-cystic fibrosis bronchiectasis in children with a focus on the role of long-term treatment with macrolides.

PMID: 30652424 | Indexed in PubMed

Bronchiectasis is a rare chronic airway disease arising from several respiratory and systemic diseases. The grade of evidence for specific treatment of childhood bronchiectasis unrelated to cystic fibrosis (CF) is low with very few randomized controlled trials. Treatment has been based mainly on evidence from studies in adults with non-cystic fibrosis bronchiectasis and patients with cystic fibrosis. Recently, long-term treatment with macrolides has been proposed. These molecules offer the advantage of anti-inflammatory and immunomodulatory properties in addition to their antibacterial properties. A total of three randomized double-blind placebo-controlled trials conducted in adults showed that macrolides taken for 6-12 months led to a significant reduction in exacerbation rates. Only one long-term, randomized double-blind placebo-controlled trial was conducted in the pediatric population. It showed that azithromycin administered weekly for up to 24 months reduced pulmonary exacerbations. Further randomized controlled studies are needed to determine the optimal dose and duration of treatment with macrolides. The clinical profile of children who would benefit from this treatment also needs to be determined.

Long-Term Prognosis of Asthma-Bronchiectasis Overlapped Patients: A Nationwide Population-Based Cohort Study.

PMID: 34734508 | Indexed in PubMed

Asthma and bronchiectasis are common chronic respiratory diseases, and their coexistence is frequently observed but not well investigated. Our aim was to study the effect of comorbid bronchiectasis on asthma. A propensity score-matched cohort study was conducted using the National Health Insurance Service-Health Screening Cohort database. From 2005 to 2008, 8,034 participants with asthma were weighted based on propensity scores in a 1:3 ratio with 24,099 participants without asthma. From the asthma group, 141 participants with overlapped bronchiectasis were identified, and 7,892 participants had only asthma. Clinical outcomes of acute asthma exacerbation(s) and mortality rates were compared among the study groups. The prevalence of bronchiectasis (1.7%) was 3 times higher in asthmatics than in the general population of Korea. Patients who had asthma comorbid with bronchiectasis experienced acute exacerbation(s) more frequently than non-comorbid patients (11.3% vs. 5.8%, P = 0.007). Time to the first acute exacerbation was also shorter in the asthmatics with bronchiectasis group (1,970.9 days vs. 2,479.7 days, P = 0.005). Although bronchiectasis was identified as a risk factor for acute exacerbation (adjusted odds ratio, 1.73; 95% confidence interval [CI], 1.05-2.86), there was no significant relationship between bronchiectasis and all-cause or respiratory mortality (adjusted hazard ratio [aHR], 1.17; 95% CI, 0.67-2.04 and aHR, 0.81; 95% CI, 0.11-6.08). Comorbid bronchiectasis increases asthma-related acute exacerbation, but it does not-raise the risk of all-cause or respiratory mortality. Close monitoring and accurate diagnosis of bronchiectasis are required for patients with frequent exacerbations of asthma.

Use of antibiotics in bronchiectasis.

PMID: 22023177 | Indexed in PubMed

Bronchiectasis is defined by the presence of abnormal bronchial widening and occurs as a consequence of chronic airway infection. It is an important and common cause of respiratory disease. Antibiotics are the main therapy used for the treatment of this condition. The article will review the use of antibiotics for the treatment of bronchiectasis. Antibiotics can be given as short-term therapy for exacerbations or as long-term/maintenance therapy. Antibiotics given by the inhalational route and macrolides are two relatively new classes of medication that may be useful for long-term therapy. There are significant concerns about the overuse resulting in antibiotic resistance. It should be emphasized that nearly all of the trials in the literature have only had small numbers of subjects. The data that is available describing the use of antibiotics in bronchiectasis can generally be regarded as preliminary.